

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think> To solve this , we need to use the de Broglie wavelength formula and the kinetic energy formula for an electron . The de Broglie wavelength is given by :

$$\lambda = \frac{h}{\sqrt{2mE}}$$

where (h) is Planck's constant , (m) is the mass of the electron , and (E) is the kinetic energy . The kinetic energy in electron volts (eV) can be converted to joules using $(1\text{ eV} = 1.602 \times 10^{-19}\text{ J})$.

Given :

- Kinetic energy $(E = 1\text{ , eV} = 1.602 \times 10^{-19}\text{ , J})$
- Mass of an electron $(m = 9.109 \times 10^{-31}\text{ , kg})$
- Planck's constant $(h = 6.626 \times 10^{-34}\text{ , J}\cdot\text{s})$

Substituting these values into the formula :

$$\lambda = \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19}}}$$

$$\lambda = \frac{6.626 \times 10^{-34}}{\sqrt{29.18 \times 10^{-50}}}$$

$$\lambda = \frac{6.626 \times 10^{-34}}{5.402 \times 10^{-25}}$$

$$\lambda = 1.22 \times 10^{-9}\text{ , m}$$

$$\lambda = 1.22 \times 10^{-9} \times 10^2\text{ , cm}$$

$$\lambda = 1.22 \times 10^{-7}\text{ , cm}$$

This value is closest to 1.23×10^{-7} cm .

</think><answer> 1.23×10^{-7} cm</answer>